

浙大 2021. 量子.

一. 1) 由 $eV = E = \frac{h}{\lambda}$

$$\lambda = \frac{h}{eV}$$

2) 坐标表象下, $\hat{x} = x$, $\hat{p} = -i\hbar \nabla$

$$[x, p]\psi = (\hat{x}\hat{p} - \hat{p}\hat{x})\psi = -i\hbar x \nabla \psi + i\hbar \nabla(x\psi)$$

$$= -i\hbar x \nabla \psi + i\hbar \psi + i\hbar x \nabla \psi = i\hbar \psi$$

$$[x, p] = i\hbar$$

3) 薛定谔方程: $-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + V(x) \psi(x) = E \psi(x)$ ①

取 $-x$: $-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(-x) + V(x) \psi(-x) = E \psi(-x)$ ②

① - ② $\Rightarrow \psi(x), \psi(-x)$ 线性相关

$$\psi(x) = \lambda \psi(-x) = \lambda^2 \psi(x)$$

$$\Rightarrow \lambda = \pm 1.$$

$$\psi(x) = \psi(-x), \text{偶宇称}, \quad \psi(x) = -\psi(-x), \text{奇宇称}$$

二. 1) 当 $x < -a$ 或 $x > a$ 时, $\psi(x) = 0$.

当 $-a \leq x \leq a$ 时,

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + aV_0 \delta(x) \psi(x) = E \psi(x)$$

2) $\psi(a) = 0, \psi(-a) = 0$

$$\psi'(0^+) - \psi'(0^-) = \frac{2maV_0}{\hbar^2} \psi(0).$$

3) 奇宇称, $\psi(0) = 0$

当 $x \neq 0$ 时, 方程

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) = E \psi(x)$$

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\frac{d^2}{dx^2} \psi(x) + k^2 \psi(x) = 0.$$

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & 0 < x < a \\ -Ae^{-ikx} - Be^{ikx}, & -a < x < 0 \end{cases}$$

由 $\psi(a) = \psi(-a) = 0, \psi(0^+) = \psi(0^-) = 0,$

及阶跃条件有

$$\psi(x) = A' \sin kx$$

$$\sin ka = 0 \Rightarrow ka = n\pi, n=1,2,3,\dots$$

$$\int |\psi(x)|^2 dx = 1 \Rightarrow A' = \sqrt{\frac{1}{a}}$$

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}, -a \leq x \leq a$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}, n=1,2,3,\dots$$

无限深方势阱:

$$\psi_k = \sqrt{\frac{1}{a}} \sin \frac{k\pi x}{2a}$$

$$E_k = \frac{k^2 \pi^2 \hbar^2}{8ma^2}$$

是取无限深方势阱中 2, 4, 6, ..., 2k 项.

(4) 偶宇称. $x \neq 0$ 时有 $\frac{d^2}{dx^2} \psi(x) + k^2 \psi(x) = 0$.

$$\psi(x) = \begin{cases} A \cos kx + B \sin kx, & 0 < x < a \\ A \cos kx - B \sin kx, & -a < x < 0 \end{cases}$$

$$\text{由 } \psi'(0^+) - \psi'(0^-) = \frac{2maV_0}{\hbar^2} \psi(0) \text{ 有}$$

$$Bk + Bk = \frac{2maV_0}{\hbar^2} A$$

$$B = \frac{maV_0}{\hbar^2 k} A = \lambda A.$$

由 $\psi(a) = \psi(-a) = 0$ 有

$$A \cos ka + \lambda A \sin ka = 0.$$

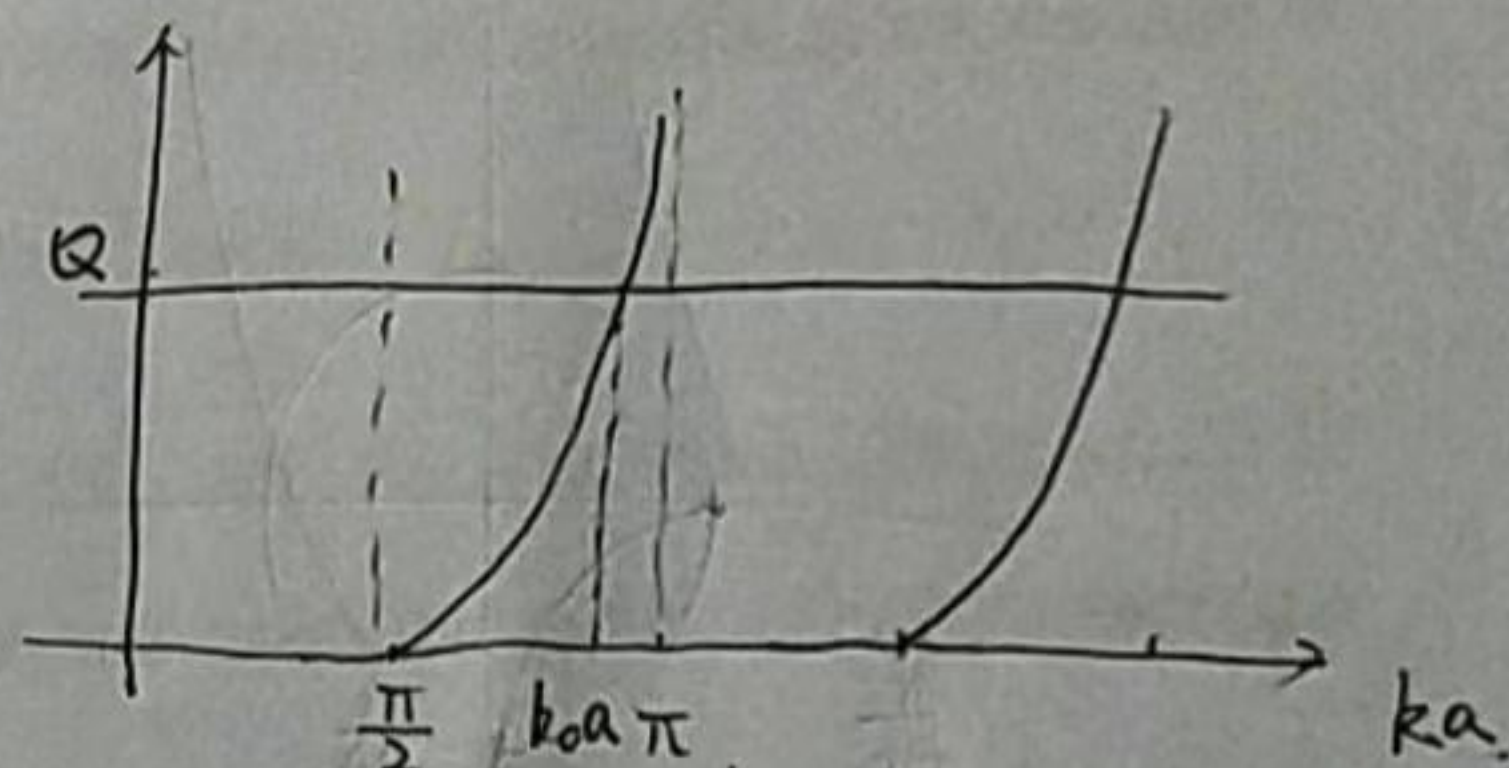
$$-k \cot ka = \frac{maV_0}{\hbar^2}$$

$$\text{令 } ka = \xi, \quad -\xi \cot \xi = \frac{maV_0}{\hbar^2} = Q$$

如图. 交点横坐标为 ka , $\frac{\pi}{2} < ka < \pi$.

$$\frac{\pi^2 \hbar^2}{8ma^2} < E_0 = \frac{\hbar^2 k_0^2}{2m} < \frac{\pi^2 \hbar^2}{2ma^2}$$

偶宇称下能级更低, 是基态.



$$\text{三、 1) } x = \sqrt{\frac{\hbar}{2m\omega}} (a_+ + a_-)$$

$$p = i\sqrt{\frac{\hbar m\omega}{2}} (a_+ - a_-)$$

$$\langle x \rangle = \int \psi_n^* x \psi_n dx$$

$$= \int \psi_n^* \sqrt{\frac{\hbar}{2m\omega}} (a_+ + a_-) \psi_n dx$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \int \psi_n^* (\sqrt{n+1} \psi_{n+1} + \sqrt{n} \psi_{n-1}) dx$$

$$= 0.$$

$$\langle p \rangle = \int \psi_n^* \hat{p} \psi_n dx$$

$$= \int \psi_n^* i\sqrt{\frac{\hbar m\omega}{2}} (a_+ - a_-) \psi_n dx$$

$$= \int i\sqrt{\frac{\hbar m\omega}{2}} \psi_n^* (\sqrt{n+1} \psi_{n+1} - \sqrt{n} \psi_{n-1}) dx$$

$$= 0.$$

$$2) \langle x^2 \rangle = \int \psi_n^* \left(\frac{\hbar}{2m\omega} \right) (a_+ + a_-)^2 \psi_n dx$$

$$= \frac{\hbar}{2m\omega} \int \psi_n^* (a_+ a_- + a_- a_+) \psi_n dx$$

$$= \frac{\hbar}{2m\omega} (2n+1)$$

$$\langle p^2 \rangle = \int \psi_n^* \left(-\frac{\hbar m\omega}{2} \right) (a_+ - a_-)^2 \psi_n dx$$

$$= \frac{\hbar m\omega}{2} \int \psi_n^* (a_+ a_- + a_- a_+) \psi_n dx$$

$$= \frac{\hbar m\omega}{2} (2n+1)$$

$$3) \sigma_x = \langle (x - \bar{x})^2 \rangle = \langle x^2 - 2x\bar{x} + \bar{x}^2 \rangle = \langle x^2 \rangle - \bar{x}^2 = \frac{\hbar}{2m\omega} (2n+1)$$

$$\sigma_p = \langle (p - \bar{p})^2 \rangle = \langle p^2 - 2p\bar{p} + \bar{p}^2 \rangle = \langle p^2 \rangle - \bar{p}^2 = \frac{\hbar m\omega}{2} (2n+1)$$

$$\sigma_x \sigma_p = \frac{\hbar^2}{4} (2n+1)^2 \geq \frac{\hbar^2}{4}$$

当且仅当 $n=0$, 即粒子处于量子态 ψ_0 时等号成立.

$$4) \langle x \rangle = \int \psi^* \sqrt{\frac{\hbar}{2m\omega}} (a_+ + a_-) \psi dx$$

$$= \int (a^* \psi_1^* + b^* \psi_2^*) \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{2} a \psi_2 + a \psi_0 + \sqrt{3} b \psi_3 + \sqrt{2} b \psi_1) dx$$

$$= \sqrt{\frac{\hbar}{2m\omega}} (\sqrt{2} a^* b + \sqrt{2} a b^*)$$

$$5) E_n = (n + \frac{1}{2}) \hbar \omega$$

$$\psi(t) = a e^{-iE_1 t/\hbar} \psi_1 + b e^{-iE_2 t/\hbar} \psi_2 = a e^{-i\frac{3}{2}\omega t} \psi_1 + b e^{-i\frac{5}{2}\omega t} \psi_2$$

$$\langle x(t) \rangle = \int \psi^*(t) x \psi(t) dx$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \int (a^* e^{i\frac{3}{2}\omega t} \psi_1^* + b^* e^{i\frac{5}{2}\omega t} \psi_2^*) [a e^{-i\frac{3}{2}\omega t} (\sqrt{2} \psi_2 + \psi_0) + b e^{-i\frac{5}{2}\omega t} (\sqrt{3} \psi_3 + \sqrt{2} \psi_1)] dx$$

$$= \sqrt{\frac{\hbar}{2m\omega}} (a^* b e^{-i\omega t} + a b^* e^{i\omega t})$$

四. 1) 氢原子能量 $E_n = -\frac{e^2}{2an^2}$

能量可能测量值为 $-\frac{e^2}{2a}$, $-\frac{e^2}{8a}$, 概率分别为 $|A|^2$, $|B|^2 + |C|^2$

2) $t=0$ 时, $\psi(0) = A\psi_{100} + B\psi_{210} + C\psi_{211}$

$\hat{L}$ 可能测量值 $0, 2\hbar^2$, 概率分别为 $|A|^2$, $|B|^2 + |C|^2$

$\hat{L}_y$ 可能 $\dots 0, \hbar, \dots$ 概率分别为 $|A|^2 + |B|^2$, $|C|^2$.

$$3) \hat{L}_x = \frac{1}{2}(\hat{L}_+ + \hat{L}_-)$$

$$\hat{L}_\pm \psi_{nlm} = \sqrt{l(l+1) - m(m\pm 1)} \hbar \psi_{nl, m\pm 1}$$

$$\psi(r, t) = A \psi_{100} e^{-iE_1 t/\hbar} + B \psi_{210} e^{-iE_2 t/\hbar} + C \psi_{211} e^{-iE_2 t/\hbar}$$

$$\hat{L}_x \psi(r, t) = \frac{\sqrt{2}}{2} \hbar B \psi_{211} e^{-iE_2 t/\hbar} + \frac{\sqrt{2}}{2} \hbar C \psi_{210} e^{-iE_2 t/\hbar}$$

L_x 可能测量值为 0, 概率 $|A|^2$.

$$\begin{array}{cc} \frac{\sqrt{2}}{2} \hbar & \frac{1}{2} (B+C)^2 \\ -\frac{\sqrt{2}}{2} \hbar & \frac{1}{2} (B-C)^2 \end{array}$$

以 ψ_{210}, ψ_{211} 为基:

$$\frac{\sqrt{2}}{2} \hbar \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \hat{L}_x.$$

$$\frac{\hbar}{2}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$-\frac{\hbar}{2}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} B e^{-iE_2 t/\hbar} \\ C e^{-iE_2 t/\hbar} \end{pmatrix}.$$

$$4) x = r \sin \theta \cos \varphi, y = r \sin \theta \sin \varphi, z = r \cos \theta$$

$$\langle x \rangle = \langle \psi(r, t) | x | \psi(r, t) \rangle$$

$$= |A|^2 \underbrace{\langle \psi_{100} | x | \psi_{100} \rangle}_{=0} + |B|^2 \underbrace{\langle \psi_{210} | x | \psi_{210} \rangle}_{=0} + |C|^2 \underbrace{\langle \psi_{211} | x | \psi_{211} \rangle}_{=0}$$

$$\frac{1}{\pi a^3} \int e^{-r/a} r^3 \sin^2 \theta \cos \varphi dr d\theta d\varphi$$

$$\int_0^{2\pi} \cos \varphi d\varphi = 0$$

$$A \int e^{-r/a} r^4 \sin^2 \theta \cos \theta \sin \varphi dr d\theta d\varphi$$

$$\int_0^{2\pi} \sin \varphi d\varphi = 0$$

$$A \int e^{-r/a} r^2 \sin \theta \cos \theta e^{i\varphi} dr d\theta d\varphi$$

$$\int_0^{2\pi} e^{i\varphi} d\varphi = 0$$

$$= 0.$$

$$\text{同理: } \langle y \rangle = 0, \langle z \rangle = 0.$$

5) 可以.

五. 1) $\vec{B} = (B, 0, 0)$

$$\hat{H} = -\vec{M}_s \cdot \vec{B} = \frac{q}{mc} \vec{S} \cdot \vec{B} = \frac{qB}{mc} \hat{S}_x = \frac{qB\hbar}{2mc} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

在 $\hat{S}_z$ 表象中,

$$t=0 \text{ 时, } \psi(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\text{设 } t \text{ 时刻, } \psi(t) = \begin{pmatrix} a(t) \\ b(t) \end{pmatrix}, \quad |a(t)|^2 + |b(t)|^2 = 1$$

由 $\hat{H}\psi(t) = i\hbar \frac{d}{dt} \psi(t)$ 得

$$\frac{qB\hbar}{2mc} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a(t) \\ b(t) \end{pmatrix} = i\hbar \frac{d}{dt} \begin{pmatrix} a(t) \\ b(t) \end{pmatrix}$$

$$\begin{cases} \frac{qB\hbar}{2mc} b(t) = i\hbar \frac{d}{dt} a(t) \\ \frac{qB\hbar}{2mc} a(t) = i\hbar \frac{d}{dt} b(t) \end{cases}$$

$$\Rightarrow \frac{d^2}{dt^2} b(t) + \left(\frac{qB}{2mc}\right)^2 b(t) = 0, \quad \text{令 } k = \frac{qB}{2mc}$$

$$b(t) = A \sin kt + B \cos kt$$

$$\text{由 } b(0) = 0 \Rightarrow B = 0 \Rightarrow b(t) = A \sin kt$$

$$a(t) = iA \cos kt$$

$$\text{由 } a(0) = 1 \Rightarrow iA = 1 \Rightarrow A = -i$$

$$\psi(t) = \begin{pmatrix} \cos kt \\ -i \sin kt \end{pmatrix}$$

$$2) \quad \hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad s_y = \frac{\hbar}{2}, \quad \psi_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \quad s_y = -\frac{\hbar}{2}, \quad \psi_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$\psi(t) = \alpha \psi_+ + \beta \psi_- = \frac{1}{\sqrt{2}} \begin{pmatrix} \alpha + \beta \\ i\alpha - i\beta \end{pmatrix} = \begin{pmatrix} \cos kt \\ -i \sin kt \end{pmatrix}$$

$$\Rightarrow \alpha = \frac{\sqrt{2}}{2} (\cos kt - \sin kt), \quad \beta = \frac{\sqrt{2}}{2} (\cos kt + \sin kt)$$

自旋 y 分量测得 $\frac{\hbar}{2}$ 概率为 $\frac{1}{2} (1 - \sin 2kt)$

$$-\frac{\hbar}{2} \quad \dots \quad \frac{1}{2} (1 + \sin 2kt)$$

$$3) \text{ 反向时, } |\cos kt| = 0, \quad |\sin kt| = 1 \Rightarrow kt = (n + \frac{1}{2})\pi, \quad n = 0, 1, 2, \dots$$

$$\text{第一次反向, } kT = \frac{\pi}{2} \Rightarrow T = \frac{\pi mc}{qB}$$

~~(1) $t \rightarrow \infty$ 时刻, $\psi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$~~

第一激发态:

$$E_1^{(1)} = \int \psi^* H' \psi dx$$

$$= \int \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) + \psi_2(x_1) \psi_1(x_2)]^* (-aV_0) \delta(x_1 - x_2) \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) + \psi_2(x_1) \psi_1(x_2)] dx$$

$$= -aV_0 \int [\psi_1(x) \psi_2(x)]^* [\psi_1(x) \psi_2(x)] dx$$

$$= -aV_0.$$

六. 1) 一维无限深势阱

$$\psi_n = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}, \quad n=1, 2, 3, \dots$$

粒子自旋为0, 玻色子.

基态, $\psi = \psi_1(x_1) \psi_1(x_2)$, $E_0 = E_{11} = \frac{\pi^2 \hbar^2}{ma^2}$

第一激发态: $\psi = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) + \psi_2(x_1) \psi_1(x_2)]$

$$E_1 = E_{12} = \frac{5\pi^2 \hbar^2}{2ma^2}$$

2) 自旋 $\frac{1}{2}$, 费米子.

基态, $\psi = \psi_1(x_1) \psi_1(x_2) |00\rangle$

$$|00\rangle = \frac{1}{\sqrt{2}} [\alpha(1) \beta(2) - \beta(1) \alpha(2)]$$

$$E_0 = E_{11} = \frac{\pi^2 \hbar^2}{ma^2}$$

第一激发态

$$\psi = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) + \psi_2(x_1) \psi_1(x_2)] |00\rangle$$

$$\psi = \frac{1}{\sqrt{2}} [\psi_1(x_1) \psi_2(x_2) - \psi_2(x_1) \psi_1(x_2)] \begin{cases} |11\rangle \\ |10\rangle \\ |1-1\rangle \end{cases}$$

$$E_1 = E_{12} = \frac{5\pi^2 \hbar^2}{2ma^2}.$$

3). $H' = -aV_0 \delta(x_2 - x_1).$

基态: $E_0^{(1)} = \int \psi^* H' \psi dx$

$$= \int \psi_1^*(x_1) \psi_1^*(x_2) (-aV_0) \delta(x_2 - x_1) \psi_1(x_1) \psi_1(x_2) dx$$

$$= -aV_0 \int_0^a (\psi_1^*)^2 (\psi_1)^2 dx$$

$$= -aV_0 \int_0^a \left(\frac{2}{a}\right)^2 \sin^4 \frac{\pi x}{a} dx$$

$$= -aV_0 \cdot \left(\frac{2}{a}\right)^2 \cdot \frac{3}{8} a = -\frac{3}{4} V_0.$$